# Inverted MAML Card 1: Round 2 Thermal Bit-Slicing Report

**Hardware:** Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.035, L2=1536, Mom=0.85, Clamp=±75 LSB

## 1. Round 2 Summary & Cumulative Comparison (R1–R2)

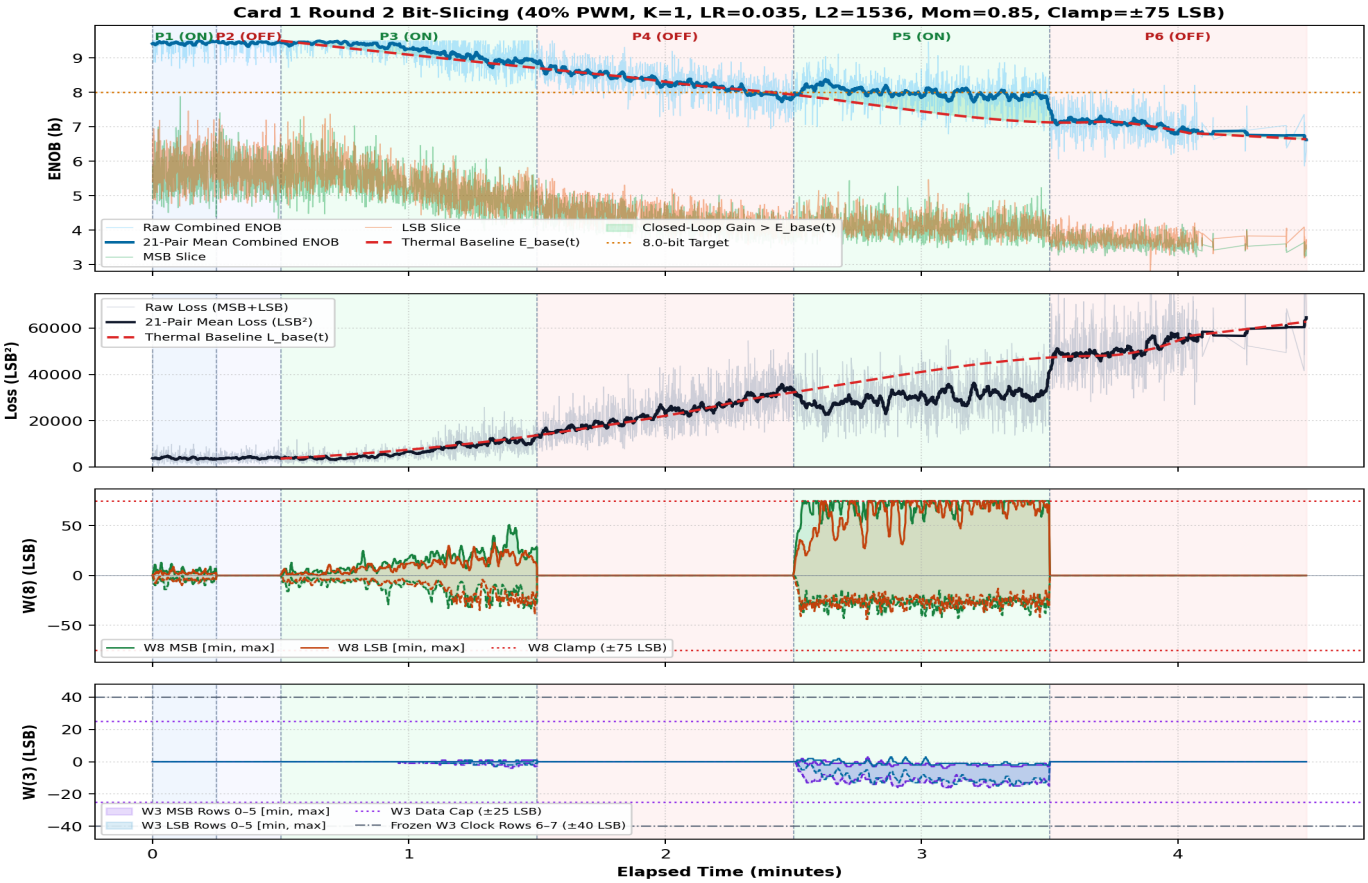
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 2 Detrended Trend Metrics:** Lift vs E_base = **+0.50 b**, Loss Ratio = **1.38x**, Climb = **+0.82 b**, Jitter = **0.29 b**, Recovery = **0.96**, Score = **+2.004**.

| Rnd | LR | L2 | Mom | Clamp | Lift | LossR | Climb | Jitt | Recov | Score |
|---|---|---|---|---|---|---|---|---|---|---|
| R1 | 0.03 | 1024 | 0.9 | ±75 | +0.66b | 1.57x | +0.74b | 0.33b | 0.67 | +1.957 |
| R2* | 0.035 | 1536 | 0.85 | ±75 | +0.50b | 1.38x | +0.82b | 0.29b | 0.96 | +2.004 |

## 2. Phase-by-Phase Telemetry (4,013 Pairs)

| Phase | Regime | N | Start->Late (Mean) | Δ vs E_base | Loss Start->Late | W8 Pk [M,L] | W3 Pk | W8 nz |
|---|---|---|---|---|---|---|---|---|
| P1 | Cold / Regler ON | 331 | 9.41->9.44 (9.43b) | - | 3743->3608 | [14,10] | 0 | 330 |
| P2 | Cold / Regler OFF | 309 | 9.45->9.45 (9.43b) | - | 3180->3834 | [0,0] | 0 | 0 |
| P3 | Warm / Regler ON | 1084 | 9.45->8.96 (9.23b) | +0.12 b | 3702->10297 | [51,39] | 4 | 1083 |
| P4 | Warm / Regler OFF | 952 | 8.92->8.03 (8.31b) | -0.03 b | 12876->29416 | [0,0] | 0 | 0 |
| P5 | Warm / Regler ON | 850 | 7.93->7.93 (8.00b) | +0.72 b | 31966->31202 | [75,75] | 16 | 850 |
| P6 | Warm / Regler OFF | 487 | 7.11->6.90 (7.06b) | +0.03 b | 49599->55672 | [0,0] | 0 | 0 |



## 3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 2

• **Full 84 W(8) + 96 W(3) Weight Coverage:** Displays the signed `[min, max]` excursion envelopes of all 42 `W(8) MSB`, 42 `W(8) LSB`, and 72 `W(3)` active data synapses (capped at ±25 LSB), plus the 24 frozen `W(3)` clock staircase weights in rows 6–7 (`[-40..+40]` & `[+40..-40]` LSB). Full per-synapse 180-trace trajectories are logged natively from Round 4 onward.

**Round 2 W(8) & W(3) Bank Coverage (84 W8 + 96 W3 Synapses; Clamp=±75, M3 Cap=±25)**

